Panskura Banamali College (Autonomous)

B.Sc.(H) 5th Semester Examination 2022 (Online mode)

Subject : BCA **Paper: CC11P**

Paper Name : Analysis of Algorithm & Graph Algorithm
F.M. : 20 **Time : 2 Hours**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020UG009967, paper_code : CC11P , then file name will be :: 2020UG009967- CC11P.pdf for question 'A' and 2020UG009967- CC11P.mp3 for question 'B']

A. Answer any one question: **1 x12 = 12**

1) Write a program to implement merge sort .
2) Write a program to implement insertion sort .
3) Write a program to implement linear search.
4) Write a program to implement quick sort .
5) Write a program to implement binary search with recursion.
6) Write a program to implement matrix multiplication with recursion .
7) Write a program to implement Tower of Hanoi probelm .
8) Write a program to implement DFS.

Marks distribution:

a. Problem Definition - 1 Marks
b. Algorithm/Flow Chart/ Pseudo-code– 5 Marks
c. Program listing/Coding - 3 Marks
d. Output/Result with discussion- 3 Marks

B) Answer any four questions: **4 x2 = 8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is time complexity ?
2. Describe Advantage of insertion sort.
3. What is Tower of Hanoi problem ?
4. What is linear search ?
5. What is binary search with recursion ?
6. What is insertion sort ?
7. What is hashing?
8. What is Big-O notation?
9. What is minimum spanning tree ?
10. What is dynamic programming?

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